HIGH-WINDING STATIONARY SOLUTIONS AND SINGULAR EXITS IN THE VAN DER POL REACTION–DIFFUSION SYSTEM

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ABSTRACT. We consider stationary solutions of the one-dimensional van der Pol reaction–diffusion system in the positive-diffusion cusp

$$d = r^4, \qquad a = 1 + \sqrt{\epsilon}\, r^3 a_2, \qquad r > 0.$$

The singular reversible folded saddle-node of type II (RFSN-II) reduction provides a reversible Hamiltonian saddle-focus core, but it does not determine the positive-diffusion future of the two exit gates leaving that compact core. Under Hypothesis H, we construct both continuations in the full stationary equations. One reaches a pole at finite spatial distance and has a Laurent–logarithmic action finite part. The other lies on a normally expanding future-staying hypersurface, reaches infinity only at infinite spatial distance, and has renormalized length and action. A resolved central–outer matching argument identifies the algebraic-directed gate with this hypersurface. Combining these constructions with the publicly archived fixed-system theorem and the compact-family clauses in Hypothesis H gives, uniformly on an existential compact positive annulus, an exhaustive return–exit partition and a countable symbolic coding of the trapped set. Periodic and nonperiodic bi-infinite return itineraries, and finite return words with the selected homoclinic endpoints, yield spatially periodic, localized multipulse, and bounded aperiodic stationary PDE profiles. No temporal stability, Turing selection, or canard identification is asserted.

1. INTRODUCTION AND MAIN RESULT

1.1. The physical stationary Hamiltonian.

Consider

$$\begin{aligned} u_t &= v - f(u) + d u_{\mathsf{xx}}, \\ v_t &= \epsilon(a-u) + v_{\mathsf{xx}}, \qquad f(u) = \frac{1}{3}u^3 - u, \end{aligned} \qquad \mathsf{x} \in \mathbb{R}, \tag{1.1}$$

where $d > 0$ and $\epsilon > 0$. Its homogeneous state is $(u, v) = (a, f(a))$. Write $\delta = \sqrt{d}$ and introduce

$$\delta u_{\mathsf{x}} = p, \qquad \delta p_{\mathsf{x}} = f(u) - v, \qquad v_{\mathsf{x}} = q, \qquad q_{\mathsf{x}} = \epsilon(u - a). \tag{1.2}$$

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Stationary profiles of (1.1) are precisely the complete spatial orbits of (1.2). With

$$\begin{aligned} F(u) &= \frac{1}{12}u^4 - \frac{1}{2}u^2, \\ \mathcal{G}_\delta &= \frac{1}{2}(\epsilon p^2 - q^2) - \epsilon\{F(u) + (a-u)v\}, \\ \lambda_\delta &= \epsilon p\, du - \delta^{-1} q\, dv, \end{aligned} \tag{1.3}$$

direct differentiation gives $d\mathcal{G}_\delta/d\mathsf{x} = 0$. Moreover, if $\omega_\delta = d\lambda_\delta$, then

$$\iota_{X_\mathsf{x}}\omega_\delta = dH_{\mathrm{ph}}, \qquad H_{\mathrm{ph}} = -\frac{\mathcal{G}_\delta - \mathcal{G}_\delta(O)}{\delta}, \qquad O = (a, 0, f(a), 0). \tag{1.4}$$

The reverser

$$\mathcal{R}(u,p,v,q) = (u,-p,v,-q) \tag{1.5}$$

is anti-symplectic. Thus the zero-energy stationary spatial problem is a reversible exact Hamiltonian system. The sign and clock in (1.4) matter for every action formula below.

1.2. **The cusp and the two terminal geometries.** Fix

$$0 < \epsilon_- < \epsilon_+ < \infty, \qquad A > 0, \tag{1.6}$$

and parameterize the positive cusp by

$$d = r^4, \qquad \delta = r^2, \qquad a = 1 + \sqrt{\epsilon}\, r^3 a_2, \qquad |a_2| \le A. \tag{1.7}$$

For small r, the homogeneous state is a saddle-focus for the spatial system. A source point that spends a long spatial time near this equilibrium makes many rotations before meeting the global event block. It may return to the source, meet a finite auxiliary boundary, enter a pole at which $u \to +\infty$ at a finite physical position, or enter an algebraic channel that reaches $u = +\infty$ only as $\mathsf{x} \to +\infty$.

These last two outcomes are genuinely different. At the pole, the physical remaining distance $s = \mathsf{x}_\mathrm{b} - \mathsf{x}$ is the natural boundary variable and the action requires a Laurent–logarithmic subtraction. At the algebraic end, $z = u^{-1}$ is the boundary defining function, physical length diverges, and both length and action are normalized against a complete reference tail.

Classical results of Devaney, Lerman, and Härterich describe recurrence, symbolic dynamics, and multipulse homoclinics near transverse saddle-focus homoclinic orbits [1, 5, 3]. Vo, Doelman, and Kaper identified the singular RFSN-II geometry and the Turing–canard regime for (1.1) [8]. The abstract first-hit theorem of [7] gives an exhaustive high-winding return–exit relation and singular action finite parts when a compact saddle-focus block and its terminal geometries meet specified hypotheses. None of these results determines the positive-diffusion future of the two van der Pol gates. At $d > 0$ the pole has a new dominant balance, while the algebraic branch must cross a degenerating central–outer transition. We construct both continuations and return the abstract relation to the original PDE. Figure 1 summarizes the geometry.

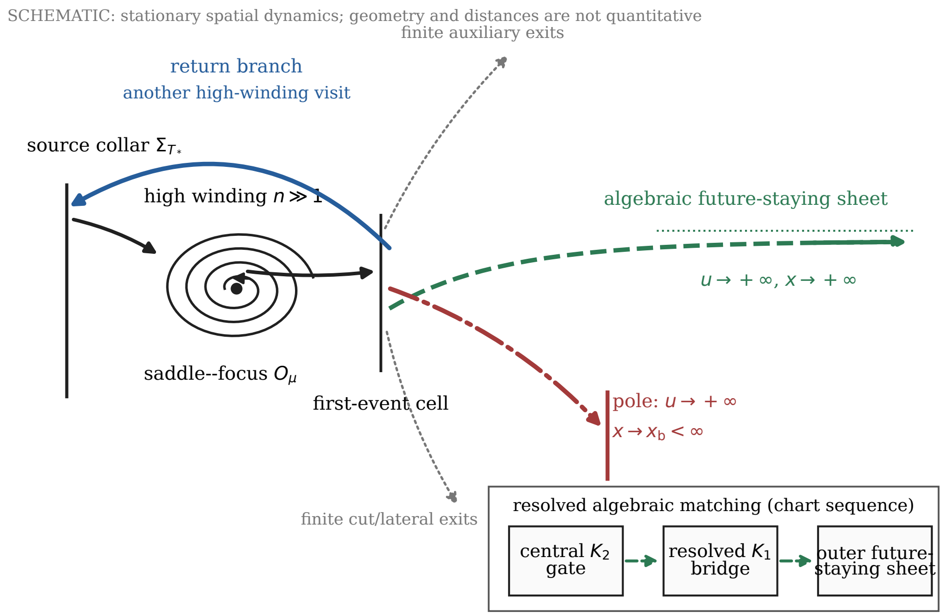

FIGURE 1. Schematic stationary spatial geometry. A point in the physical source collar makes a high-winding saddle-focus passage and then returns, reaches the finite-distance pole, reaches the infinite-distance algebraic end, or meets a named finite boundary. The inset identifies the central K_2 gate, resolved K_1 bridge, and outer future-staying sheet used to attach the algebraic event. Curve placement and distances are qualitative. The figure does not represent temporal stability, Turing selection, or canard tracking.

1.3. **Publicly archived theory and standing input.** Three logically distinct layers enter the argument. The fixed-system high-winding theorem, the validated core homoclinic and event block, and the fixed-system action and coding modules are publicly archived in [7, Theorems 2.9 and 6.3 and Corollaries 7.1–7.2]. The state-C^4 comparison sheet with its fourth rate gap, the exact Jost row and exchange coefficient, and the compact-parameter and mixed-family clauses absent from that focused preprint are imported from the exact frozen source snapshot [6]. The two positive-diffusion terminal geometries are not imported: the pole, the outer future-staying hypersurface, their central–outer matching, and both terminal finite parts are proved below. The supplement identifies the two source layers by DOI, commit, and hash, and gives the certificate and replay locations.

Here is the geometric dictionary needed to read the formal hypothesis. A small punctured source section carries a signed transverse action ν; its sign is $\sigma = \operatorname{sign} \nu$, and n records the number of local saddle-focus rotations. The central event block is a compact flow region whose first-hit faces represent a return, the local stable cut, named lateral boundaries, and two finite terminal gates. An anchor is a certified interior source phase whose first hit lies in a specified gate; a carrier is the cooriented event face containing that hit. At reader level, ($\mathrm{H}_{\mathrm{ret}}$) says the following: if this compact geometry persists and the two gates are connected to genuine terminal potentials, then every sufficiently high winding has all four source/target sign returns, an exhaustive first-event partition, and the countable-shift coding stated in Theorem 1.2. The detailed conditions below are the quantitative version of that implication.

Throughout the paper, a mixed C^k estimate is taken on the displayed fixed state domains and means a uniform bound for all state/parameter derivatives $D_Z^i D_\mu^j$ with $i+j \le k$ and $j \le 2$. Weighted saddle estimates may also contain any fixed number of logarithmic derivatives $D_{\log\nu} = \nu\partial_\nu$.

Hypothesis 1.1 (RFSN-II core and first-hit hypotheses)**.** The following data and analytic implication hold at the singular core and in the associated fixed-system construction.

($\mathrm{H}_{\mathrm{core}}$) The reversible exact Hamiltonian core

$$U' = P, \qquad P' = -U^2 - V, \qquad V' = Q, \qquad Q' = U \tag{1.8}$$

has the selected symmetric homoclinic orbit, transverse modulo the flow in the regular zero-energy hypersurface. Its true unstable source graph and one compact physical event block have certified first-hit assignments, clean and neat incidences, exhaustive connected sign strata, fixed corner priorities, and strictly positive rank, speed, separation, and ordering margins. The block contains distinct homoclinic, algebraic-directed, and pole-directed anchors. Local uniqueness of the homoclinic is required only in the frozen shooting tube. More specifically, on the nonempty pole phase interval $I_{\mathrm{p}} = [-0.2, 0.2]$, put

$$x = -U, \quad y = -P, \quad z_{\mathrm{c}} = -V, \quad \zeta = -Q, \quad D = \tfrac{1}{2}x^2 - z_{\mathrm{c}}, \quad K = xy - \zeta.$$

Every frozen source orbit in I_{p} has a strict first hit of $x = 10$ in $\{y > 0, D > 0, K > 0\}$, with uniform positive hit-speed, cone-entry, and earlier-event margins. The terms clean, neat, sign stratum, and corner priority have exactly the meaning of the adapted first-hit block in [7, Definition 2.5 and Proposition 2.8]; their fixed-core realization is part of [7, Theorem 6.3].

($\mathrm{H}_{\mathrm{Jost}}$) On a fixed central section $h_{\mathrm{c}} = 0$, the singular comparison sheet is state-C^4, with the fourth rate gap required by the Jost construction, and has parameter regularity C^2 and a defining function g_{a}. If $y_0(\phi, t) = \Phi_0^t(S_0(\phi))$ is the source flight to its selected intersection, then

$$s_0 = dh_{\mathrm{c}}(F_0)(y_0) \neq 0, \qquad \chi_0 = Dg_{\mathrm{a}}(y_0)D_\phi y_0 \neq 0. \tag{1.9}$$

The same sheet has an endpoint-normalized adjoint/Jost row ψ and a fixed growing complement $\mathbf{u}_0$ whose directed exchange pairing is

$$\chi^0_{\mathrm{ex}} = \psi(\mathbf{u}_0) = 144\sqrt{3}. \tag{1.10}$$

($\mathrm{H}_{\mathrm{ret}}$) The frozen mixed-passage, selector, first-exit-family, covariance, and terminal-extension results are available with the following input–output contract. Their inputs are: a compact C^2 family of reversible exact saddle-foci with an exact transverse-action passage; a transverse compact homoclinic transition with a fixed limiting matching rectangle, inverse and range margins; a finite state-C^3, parameter-C^2 clean physical event arrangement with strictly positive rank, speed, buffer, and ordering margins; and terminal potentials with one spare state derivative. For

$$T = -\alpha_\mu^{-1}\log|\nu| + t_{\mu,\sigma} + R_T, \qquad \Delta = -\frac{\beta_\mu}{\alpha_\mu}\log|\nu| + b_{\mu,\sigma} + R_\Delta,$$

the remainders and their mixed two-jets are $O(|\nu|(1+|\log|\nu||))$ in $D_{\log\nu}$. The event input includes every competing physical event and every pairwise event-time difference: an empty tie class has a positive gap, while a nonempty tie has nonzero conormal and a fixed priority. Under these inputs the frozen results produce one uniform winding selector, completed contracting cross forms, an exhaustive stratified first-event decomposition, exact section/gauge covariance, and the two-vertex countable terminal coding theorem with mixed two-jet estimates.

Hypothesis 1.1 imports neither positive-parameter end. The pole, outer future-staying hypersurface, their matching, and both terminal finite parts are all constructed below from (1.2). The invariant-manifold arguments use the classical compact theories of [2, 4]; the relative-with-corners and finite-atlas specializations needed here are given explicitly in the proofs below.

1.4. Main theorem.

Theorem 1.2 (Positive-diffusion return–exit theorem). *Assume Hypothesis 1.1. For the constants in* (1.6)*, there exists $r_{\mathrm{V}} > 0$ such that all of the following statements hold uniformly on the nonempty compact positive annulus*

$$\mathcal{P}_{\mathrm{V}} = \left[\frac{1}{2}r_{\mathrm{V}}, r_{\mathrm{V}}\right] \times [-A, A] \times [\epsilon_-, \epsilon_+], \qquad \mu = (r, a_2, \epsilon), \tag{1.11}$$

with physical parameters given by (1.7)*.*

(1) *The homogeneous state is a uniform saddle-focus, and the selected symmetric zero-energy homoclinic, its local stable and unstable manifolds, its exact local passage, and the compact central first-event block continue with two derivatives in μ in the blown-up parameter coordinates.*

(2) *There are a physical source collar Σ_{T_*}, a finite compatible marked cover, and an integer N such that every point with local winding $n \geq N$ has exactly one compact first event: return, stable cut, a named finite lateral event, or a terminal stratum labelled by the pole or algebraic outcome. Every point in either labelled terminal stratum has the stated subsequent noncompact future. The connected strata form an exhaustive stratified decomposition with no unnamed residual component. For each current sign σ and every $n \geq N$, both return target signs σ' occur. For every $n \geq N$, the selected anchored pole component and the selected anchored algebraic component are nonempty, with their fixed source-sign and component labels. No assertion is made that every terminal or lateral component label is nonempty for both source signs. On chart overlaps, the physical first event agrees, while raw winding integers may differ by a bounded recoding.*

(3) *Every pole-labelled terminal orbit reaches a finite physical position x_{b} and has a mixed-C^2 renormalized action*

$$\mathcal{A}_{\mathrm{p}}^{\mathrm{ren}} = \lim_{s \downarrow 0} \left[\int_{\mathsf{x}_0}^{\mathsf{x}_{\mathrm{b}} - s} \lambda_\delta - 2\epsilon\delta^3 s^{-3} + 2\epsilon\delta s^{-1} - \sqrt{6}\,\epsilon Z_0 \log s \right] \tag{1.12}$$

where Z_0 is the finite boundary label of the pole orbit and x_0 is the fixed source cut in the chosen marking; moving it changes the terminal potential by exactly the intervening finite branch action. Every algebraic-labelled terminal orbit reaches $u = +\infty$ only at infinite physical distance. In the coordinate $Q = z^{-2}$ it has mixed-C^2 renormalized length and action

$$\begin{aligned} L_{\mathrm{a}}^{\mathrm{ren}} &= \lim_{Q \to \infty} \{ L_{\mathrm{source} \to Q} - C_{\mathsf{x},\mu}(Q) \}, \\ \mathcal{A}_{\mathrm{a}}^{\mathrm{ren}} &= \lim_{Q \to \infty} \left\{ \int_{\mathrm{source} \to Q} \lambda_\delta - C_{\mathsf{A},\mu}(Q) \right\}, \end{aligned} \tag{1.13}$$

where the counterterms are complete reference-tail integrals and satisfy

$$\begin{aligned} C_{\mathsf{x},\mu}(Q) &= q_*^{-1} Q^{1/2} + O_{C^2_\mu}(\log Q), \\ C_{\mathsf{A},\mu}(Q) &= -\frac{q_*}{5\delta} Q^{5/2} + O_{C^2_\mu}(Q^2), \qquad q_* = \sqrt{\epsilon/2}. \end{aligned} \tag{1.14}$$

All finite branch actions and the two terminal potentials obey exact first-hit composition; closed actions are independent of admissible cuts, exact gauges, event-free section slides, and the local marking.

(4) *Fix $\mu \in \mathcal{P}_{\mathrm{V}}$ and a marked chart containing it. The represented local trapped return set is conjugate to the two-vertex countable edge shift with edges*

$$(\sigma, n, \sigma'), \qquad \sigma, \sigma' \in \{+, -\}, \qquad n \geq N. \tag{1.15}$$

Primitive periodic words yield smooth spatially periodic stationary solutions of (1.1), and nonperiodic bi-infinite words yield bounded nonperiodic stationary solutions. The completed first-event relation, *with its selected unstable and stable boundary traces, realizes arbitrary finite admissible edge words as homoclinic multipulse profiles converging to* $(a, f(a))$ *as* $\mathsf{x} \to \pm\infty$.

Items 1–3 classify the entire sufficiently high-winding source collar. The pole and algebraic branches are terminal escape outcomes; they are not bounded stationary patterns. The bi-infinite trapped return set in item 4 produces the periodic and bounded aperiodic profiles. Multipulses instead come from the boundary-completed finite-word relation: they make only finitely many returns before converging to the homogeneous state in both spatial directions. The theorem is local to the displayed source collar, not a classification of the global phase space. Its parameter set (1.11) is existential: it is not the current numerical candidate box, it is not uniform down to the cusp tip $r = 0$, and its image in (d, a, ϵ) is a cusp rather than a rectangular box.

For completeness, the quantitative high-winding consequence is stated next. Let

$$c_\mu = 2ra_2 + \sqrt{\epsilon}\, r^4 a_2^2, \qquad \alpha_\mu = \frac{1}{2}\sqrt{2 + c_\mu}, \qquad \beta_\mu = \frac{1}{2}\sqrt{2 - c_\mu}. \tag{1.16}$$

For any

$$0 < \varkappa < \inf_{\mu \in \mathcal{P}_{\mathrm{V}}} \frac{2\pi\alpha_\mu}{\beta_\mu}, \tag{1.17}$$

the marked cover and N may be chosen so that its cross forms and branch potentials converge in mixed C^2 at rate $C(1+n)^3 e^{-\varkappa n}$. Fix $\mu \in \mathcal{P}_{\mathrm{V}}$ and one marked chart. For a fixed primitive cyclic word with windings $n_0, \ldots, n_{m-1}$,

$$L(\gamma_{\mathbf{n}}) = \frac{2\pi r}{\epsilon^{1/4}\beta_\mu} \sum_{j=0}^{m-1} n_j + \mathcal{L}_{\boldsymbol{\sigma},\mu} + O\Big(\max_j (1+n_j)^3 e^{-\varkappa n_j}\Big), \tag{1.18}$$

while its closed action has a finite marking-independent limit with the same error. Here μ, the word length, and the cyclic sign sequence are fixed, and the limit means $\min_j n_j \to \infty$. The estimate is not uniform when the number of composed branches tends to infinity. The notation $\boldsymbol{\sigma}$ records the fixed cyclic sequence of source and target signs, so the finite offset need not be a function of a single sign.

1.5. **Proof strategy: from finite gates to genuine exits.** The publicly archived fixed-system theorem reaches the two compact central gates; it does not identify their positive-diffusion futures. The proof has four steps. First, central continuation transports the singular event block to a compact positive-parameter family. Second, an invariant cone and a regular-singular compactification connect the pole-directed gate to genuine finite-distance blow-up and its terminal action. Third, an outer graph transform constructs the algebraic future near infinity, and a resolved K_1 bridge proves that the central algebraic gate lies on that same future-staying sheet. Finally, replacing the two finite gates inside their protected flow boxes preserves the complete first-event arrangement, so the high-winding theorem applies to the original stationary PDE. The logical chain is therefore: singular core, finite gates, genuine terminals, exhaustive first events, and finally stationary PDE profiles. Here K_2 is the central chart containing the finite algebraic gate, whereas K_1 resolves the intermediate corner through which that gate is matched to the outer sheet. Sections 2 and 3 remove the two model-specific obstructions; Section 4 completes the physical first-event relation.

2. Central continuation and the finite-distance pole

The central scaling transfers the validated singular core to a compact positive-parameter family. Its immediate output is a pair of finite exit gates; identifying their noncompact futures requires separate arguments.

2.1. **Exact central scaling.**

The cusp scaling is fixed before applying the imported core. Put

$$\begin{aligned} u &= a - \sqrt{\epsilon}\, r^2 U, \quad p = -\epsilon^{3/4} r^3 P, \\ v &= f(a) - \epsilon r^4 V, \quad q = -\epsilon^{5/4} r^3 Q, \quad \xi = \epsilon^{1/4}\mathsf{x}/r. \end{aligned} \tag{2.1}$$

This sends the homogeneous state to the origin. Direct substitution, with no remainder, gives

$$\begin{aligned} U' &= P, \\ P' &= c_\mu U - V - (1 + \sqrt{\epsilon}\, r^3 a_2) U^2 + \frac{\sqrt{\epsilon}}{3} r^2 U^3, \\ V' &= Q, \qquad Q' = U, \end{aligned} \tag{2.2}$$

where the prime denotes $d/d\xi$ and c_μ is defined in (1.16). The parameter-independent primitive and shifted Hamiltonian are

$$\begin{aligned} \lambda &= P\, dU - Q\, dV, \\ \widehat{H}_\mu &= \frac{1}{2}(Q^2 - P^2) - UV + \frac{c_\mu}{2} U^2 - \frac{1 + \sqrt{\epsilon} r^3 a_2}{3} U^3 + \frac{\sqrt{\epsilon}}{12} r^2 U^4, \\ \iota_{F_\mu} d\lambda &= d\widehat{H}_\mu. \end{aligned} \tag{2.3}$$

The exact physical conversions are

$$d\mathsf{x} = r\epsilon^{-1/4}\, d\xi, \qquad \lambda_\delta = \epsilon^{9/4} r^5 \lambda, \qquad H_{\mathrm{ph}} = \epsilon^{5/2} r^4 \widehat{H}_\mu. \tag{2.4}$$

At $r = 0$, (2.2) is exactly (1.8). Its characteristic polynomial at the origin is $\zeta^4 - c_\mu \zeta^2 + 1$, and hence its spectrum is $\{\pm\alpha_\mu \pm i\beta_\mu\}$.

Proposition 2.1 (Central continuation)**.** *Under* ($\mathrm{H}_{\mathrm{core}}$) *and the mixed-passage clause of* ($\mathrm{H}_{\mathrm{ret}}$)*, after decreasing the upper r-radius, the origin, the selected symmetric homoclinic, and the compact central first-event block continue on the closed blown-up parameter wedge. The local invariant manifolds and the homoclinic have two parameter derivatives with uniform exponential tails. A finite parameter cover carries reversible exact saddle coordinates. If*

ν is the signed transverse action, the passage time and Kato-oriented phase satisfy

$$\begin{aligned} T_{\mu,\sigma}(\nu) &= -\alpha_\mu^{-1}\log|\nu| + t_{\mu,\sigma} + O_{\mathrm{wl}}(|\nu|), \\ \Delta_{\mu,\sigma}(\nu) &= -\frac{\beta_\mu}{\alpha_\mu}\log|\nu| + b_{\mu,\sigma} + O_{\mathrm{wl}}(|\nu|), \end{aligned} \tag{2.5}$$

where $O_{\mathrm{wl}}(|\nu|)$ denotes $O(|\nu|(1+|\log|\nu||))$ after any two parameter derivatives and any fixed number of logarithmic derivatives $\nu\partial_\nu$. The continued event block remains clean, neat, componentwise exhaustive, and separated by uniform positive physical first-hit margins.

Proof. On fixed compact state sets, (2.2) is a smooth C^2 parameter family converging to (1.8). The saddle-focus rates stay uniformly away from zero. Parameter-dependent graph transforms give fixed domain stable and unstable graphs; their Lyapunov–Perron equations may be differentiated twice because the exponential dichotomy is uniform. In a weighted full-line space, the reversible matching equation for the selected core orbit has invertible derivative by the certified transversality in ($\mathrm{H}_{\mathrm{core}}$). The implicit-function theorem therefore gives the homoclinic and its two-derivative exponential tails.

The exact local passage follows from the reversible analytic saddle normal form in ($\mathrm{H}_{\mathrm{ret}}$) after tangent normalization to one Kato-transported oriented spectral frame. Direct normal-form quadrature on the radial sections fixes the minus sign in the phase law in (2.5); the transverse action itself is unchanged. Finally, all compact first-hit speeds, active conormal ranks, empty incidences, earlier-event exclusions, and ordering inequalities in ($\mathrm{H}_{\mathrm{core}}$) are strict on compact sets. Smooth first-hit dependence and a boundary-preserving controlled isotopy retain the complete labelled incidence complex. This proves the proposition. ☐

The two end labels in Proposition 2.1 are still finite gates. Nothing in compact first-hit stability implies that an orbit subsequently reaches a noncompact end.

2.2. **Invariant cone and finite physical blow-up.** For the pole-directed branch write

$$x = -U, \qquad y = -P, \qquad z = -V, \qquad \zeta = -Q, \tag{2.6}$$

and set

$$D = \frac{1}{2}x^2 - z, \qquad K = xy - \zeta, \qquad \mathcal{K}_\mu = \{x \ge 10,\ y > 0,\ D \ge 0,\ K \ge 0\}. \tag{2.7}$$

The exact field gives

$$\begin{aligned} x' &= y, & D' &= K, \\ y' &= D + (B - \tfrac{1}{2})x^2 + c_\mu x + bx^3, & K' &= y^2 + xy' - x. \end{aligned} \tag{2.8}$$

where $B = 1 + \sqrt{\epsilon} r^3 a_2$ and $b = \sqrt{\epsilon} r^2/3 > 0$. On the final small positive annulus, $B \geq 3/4$ and $|c_\mu| \leq 1$. Hence the faces of $\mathcal{K}_\mu$ point inward and, for $x \geq 10$,

$$y' > 0, \qquad K' > 0, \qquad y\frac{dy}{dx} \geq bx^3. \tag{2.9}$$

The certified core source window has a strict first hit of $x = 10$ in the interior of this cone. Proposition 2.1 and a short event-free flowbox slide retain that literal first hit for positive parameters. From (2.9),

$$y(x)^2 \geq y(10)^2 + \frac{b}{2}(x^4 - 10^4), \qquad \xi_{\mathrm{b}} - \xi = \int_x^\infty \frac{dX}{y(X)} < \infty. \tag{2.10}$$

Thus the pole is reached in finite central and physical spatial time. On the zero-energy level the exact identity is

$$y^2 = \zeta^2 - 2xz + c_\mu x^2 + \frac{2B}{3}x^3 + \frac{b}{2}x^4. \tag{2.11}$$

The lower bound in (2.10) gives

$$\frac{d\zeta}{dx} = \frac{x}{y} = O(x^{-1}), \qquad \frac{dz}{dx} = \frac{\zeta}{y} = O((1 + \log x)x^{-2}).$$

Hence $z \to z_\infty$, $\zeta = O(1 + \log x)$, and (2.11) closes the uniform limits

$$\frac{y}{x^2} \to \sqrt{b/2}, \qquad \zeta - \sqrt{2/b}\log x \to \zeta_\infty, \qquad z_\infty - z = O((1 + \log x)x^{-1}). \tag{2.12}$$

One further integration of $dx/d\xi = y$ and the exact physical inverse of (2.1) give the balance below. In particular, the two end labels range in a compact set on the transported source window.

2.3. **Regular-singular compactification and pole action.** Let $s = \mathsf{x}_{\mathrm{b}} - \mathsf{x}$ and $\ell = \sqrt{6}\,\delta$. The sharp balance obtained from the conserved energy and (2.9) is

$$\frac{su}{\ell} \to 1, \qquad \frac{s^2 p}{\ell\delta} \to 1, \qquad q + \ell\epsilon\log s \to W_0, \qquad v - \ell\epsilon s\log s \to Z_0. \tag{2.13}$$

Define

$$X = su/\ell, \qquad Y = s^2 p/(\ell\delta), \qquad W = q + \ell\epsilon\log s, \qquad Z = v - \ell\epsilon s\log s, \tag{2.14}$$

and use $\dot{} = s\,d/d\mathsf{x}$. Substitution in (1.2) gives the exact C^3 regular-singular field

$$\begin{aligned}
&\dot{s} = -s, && \dot{X} = -X + Y,\\
&\dot{Y} = -2Y + 2X^3 - s^2\delta^{-2}X - s^3(\ell\delta^2)^{-1}Z - \epsilon\delta^{-2}s^4\log s, &&\\
&\dot{W} = \ell\epsilon(X - 1) - a\epsilon s, && \dot{Z} = s(W + \ell\epsilon).
\end{aligned} \tag{2.15}$$

At $s = 0$ it has the two-dimensional equilibrium manifold $X = Y = 1$, parameterized by (Z_0, W_0). Its normal spectrum is $\{-1, -4, +1\}$, while the two label directions have eigenvalue zero. The stable pole family therefore has normalized power spectrum

$$\{-1, 0, 0, 1, 4\}. \tag{2.16}$$

Proposition 2.2 (Positive pole terminal). *Under Hypothesis 1.1, after decreasing only r_{V} if necessary and retaining the full $[-A, A] \times [\epsilon_-, \epsilon_+]$ factors, the transported pole source window enters an open pole basin and reaches $u = +\infty$ at a finite physical position. Each pole orbit has unique labels (Z_0, W_0, c_4) and a polyhomogeneous expansion with admissible powers 1 and 4, including the forced $s \log s$ and $s^4(\log s)^2$ terms. On the fixed zero-energy level, c_4 is determined uniquely because*

$$\partial_{c_4} \mathcal{G}_\delta = -30\epsilon\delta^4 \neq 0. \tag{2.17}$$

The action finite part (1.12) exists with two mixed parameter derivatives and satisfies exact moving-cut additivity. On every fixed compact pole-entry chart with coordinates $\mathbf{e}_{\mathrm{p}}$, let L_{p} be the remaining physical length from the fixed entry cut to x_{b}. Then L_{p} and the renormalized action have uniformly bounded derivatives

$$\sup_{i+j\le 3,\ j\le 2} \left(\left\| D^i_{\mathbf{e}_{\mathrm{p}}} D^j_\mu L_{\mathrm{p}} \right\| + \left\| D^i_{\mathbf{e}_{\mathrm{p}}} D^j_\mu \mathcal{A}^{\mathrm{ren}}_{\mathrm{p}} \right\| \right) < \infty. \tag{2.18}$$

This is one spare entry derivative; no additional derivative in the singular variable s is asserted.

Proof. Normal hyperbolicity of the compact label rectangles in (2.15) gives a four-dimensional local stable set: two label directions and two stable fibers. The uniform limits (2.13) and (2.12) show that every transported source orbit converges to the interior of one common compact label rectangle. The defining staying-set characterization of the local stable manifold therefore puts the entire window in that stable set. Eliminating Y yields the scalar Fuchsian equation

$$(s\partial_s - 4)(s\partial_s + 1)(X - 1) = 6(X-1)^2 + 2(X-1)^3 - \frac{s^2}{\delta^2} X - \frac{s^3}{\ell\delta^2} Z - \frac{\epsilon}{\delta^2} s^4 \log s. \tag{2.19}$$

The nonresonant powers two and three are forced; at the indicial root four, the constant and logarithmic forcing produces $s^4 \log s$ and $s^4(\log s)^2$, with $c_4 s^4$ the free stable mode. The Green inverse of $(s\partial_s - 4)(s\partial_s + 1)$ is bounded on the conormal weight $s^5(1 + |\log s|)^2$. A uniform contraction, differentiated in the entry labels and parameters through total order three with at most two parameter derivatives, proves the expansion and the mixed regularity in (2.18). The physical inverse of (2.14) has nonzero leading Jacobian, so these are genuine local state coordinates and their image is an open pole basin. Substitution into (1.3) gives (2.17).

Finally,

$$\lambda_\delta(\partial_{\mathsf{x}}) = 6\epsilon\delta^3 s^{-4} - 2\epsilon\delta s^{-2} - \sqrt{6}\,\epsilon Z_0 s^{-1} + O_{C^2}((1 + |\log s|)^2). \tag{2.20}$$

The remainder is integrable uniformly after two mixed derivatives and after the one spare entry derivative in (2.18). Integrating (2.20) proves (1.12). Moving a finite cut merely transfers the same finite orbit integral between the central and terminal terms, proving exact additivity. □

Proposition 2.2 therefore replaces the pole-directed gate by an open basin of genuine finite-distance blow-up and supplies its terminal action. The algebraic-directed gate remains unresolved.

3. The algebraic end: outer geometry, matching, and finite parts

The algebraic terminal requires two distinct steps. We first characterize the tails that remain in a compactified outer corridor. We then prove that the algebraic-directed central gate enters precisely this future-staying hypersurface.

3.1. **A positive-parameter future-staying hypersurface.**

The algebraic channel is most transparent in the fast clock $y = \mathsf{x}/\delta$. For $u > 0$ set

$$z = u^{-1}, \qquad \pi = p, \qquad w = z\{f(u) - v\}, \qquad \chi = z^2 q, \qquad \frac{d}{d\tau} = z\frac{d}{dy}. \tag{3.1}$$

The exact compactified field is

$$\begin{aligned} \dot z &= -\pi z^3, & \dot\pi &= w, \\ \dot w &= (1 - z^2)\pi - \delta\chi - \pi w z^2, & \dot\chi &= z^2\{\epsilon\delta(1 - az) - 2\pi\chi\}. \end{aligned} \tag{3.2}$$

Diagonalize the fast normal pair by

$$h = \pi - \delta\chi, \qquad \alpha = (h + w)/2, \qquad \beta = (h - w)/2. \tag{3.3}$$

Thus $\pi = \delta\chi + \alpha + \beta$ and $w = \alpha - \beta$. The shifted energy

$$E = -\mathcal{G}_\delta + \mathcal{G}_\delta(O) = -\mathcal{G}_\delta - \epsilon F(a) \tag{3.4}$$

is a regular boundary coordinate. In the outer variables its exact energy equation is

$$\begin{aligned} \chi^2 = \frac{\epsilon}{2} - \frac{2a\epsilon}{3}z - \epsilon(2w + 1)z^2 + 2a\epsilon(w + 1)z^3 \\ + \{\epsilon\pi^2 + 2E + 2\epsilon F(a)\}z^4. \end{aligned} \tag{3.5}$$

At $z = 0$ the derivative with respect to the positive χ root is $2\sqrt{\epsilon/2} > 0$. Hence the implicit-function theorem gives

$$\chi = \mathrm{Chi}(z, E, \beta, \alpha; \mu) > 0, \qquad \mathrm{Chi}(0, E, \beta, \alpha; \mu) = q_*(\epsilon) = \sqrt{\epsilon/2}, \tag{3.6}$$

and the boundary field in (z, E, β, α) is exactly

$$(\dot z, \dot E, \dot\beta, \dot\alpha) = (0, 0, -\beta, \alpha). \tag{3.7}$$

Thus β is future stable and α future unstable. The positive cubic balance has already been absorbed in the regular root (3.6); there is no coefficient singular in r at $z = 0$.

Proposition 3.1 (Outer future-staying graph)**.** *On the final compact positive annulus, there is a fixed corridor in (z, E, β, α) whose maximal forward-staying set is a unique codimension-one graph*

$$\mathcal{W}^{\mathrm{tail}}_{\mathrm{out},\mu} = \{\alpha = \Gamma_\mu(z, E, \beta)\}. \tag{3.8}$$

It is a locally maximal C^3 manifold with corners, normally expanding in forward τ-time, third-order bunched, and has mixed regularity

$$\Gamma \in C^0_\mu C^3_{z,E,\beta} \cap C^1_\mu C^2_{z,E,\beta} \cap C^2_\mu C^1_{z,E,\beta}. \tag{3.9}$$

Every orbit on the zero-energy future sheet with $z > 0$ satisfies

$$\begin{aligned} u_{\mathsf{x}} &= q_* + O(u^{-1}), & q &= q_* u^2 + O(u), \\ v &= f(u) + O(u^{-1}), & p &= \delta q_* + O(u^{-1}). \end{aligned} \tag{3.10}$$

and reaches $z = 0$ only as $\mathsf{x} \to +\infty$.

Proof. Choose a product corridor so that $z = 0$ and the energy faces are invariant, the remaining base faces point inward, and the two α faces are strict exits. By (3.7), after decreasing the fixed outer radius, the base logarithmic norm and the two off-diagonal derivative blocks are at most ν, while the normal expansion is at least $1-\nu$. The tangent and normal quotient cocycles for one fixed time $T > 0$ then obey

$$\sup_{\mu, Z} \left\|(N^T_{\mu,Z})^{-1}\right\| \left\|A^T_{\mu,Z}\right\|^j < 1, \qquad j = 0, 1, 2, 3. \tag{3.11}$$

Extend the field across the noninvariant faces of a doubled collar while preserving the displayed face signs and quotient gaps. The time-T graph transform on sections $\alpha = \Gamma(z, E, \beta)$ is a contraction in C^3: the estimate for its jth state derivative is exactly the jth inequality in (3.11). Its fixed graph is independent of the chosen extension on the original corridor, because a forward-staying orbit there never meets the extended region. Conversely, every point off the graph acquires a growing normal component and exits through an α face, so the graph is the maximal forward-staying set. Differentiating the fixed-point equation once and twice in μ and using the same inequalities with one and two state derivatives spent gives (3.9). This is the relative-with-corners specialization of the compact section theorem [2, 4].

On the graph, $d(z^{-2})/d\tau = 2\pi$ with π uniformly positive. Bounded variation of constants removes the growing α mode and gives $\alpha, \beta = O(z^2)$. The energy root then yields $\chi = q_* + O(z)$ and $\pi = \delta q_* + O(z)$. Substitution in (3.1) proves (3.10); since $d\mathsf{x}/d\tau = \delta z$ and $z(\tau) \asymp (1+\tau)^{-1/2}$, the physical length diverges. $\square$

3.2. **Resolution of the $K_2 \to K_1 \to$ outer corner.** Proposition 3.1 does not imply that the finite algebraic-directed gate from Proposition 2.1 lies on (3.8). The gate lives in the universal central chart K_2; the future sheet is constructed in the physical outer chart; and the two meet through the entry/exit chart K_1. The overlap degenerates as $r \to 0$, so a compact-away-from-the-corner Fenichel argument does not control the two parameter derivatives needed by the return theorem.

The exact K_2–K_1 transition on the positive overlap is

$$\begin{aligned} r_1 &= r\sqrt{u_2}, & \delta_1 &= u_2^{-1}, \\ p_1 &= u_2^{-3/2}p_2, & v_1 &= u_2^{-2}v_2, \\ q_1 &= u_2^{-3/2}q_2, & a_1 &= u_2^{-3/2}a_2, \end{aligned} \tag{3.12}$$

with $dy_2 = u_2^{-1/2}dy_1$. We resolve the corner on a closed comparison family first, choose all smallness thresholds there, and only afterward freeze a positive annulus.

The outer trace remains regular at the comparison face after setting

$$A = \alpha/\delta, \quad \mathsf{B} = \beta/\delta, \quad H = \frac{E}{\epsilon^{5/2}r^6}, \quad S = \chi + A + \mathsf{B}, \quad D_{\mathrm{o}} = A - \mathsf{B}. \tag{3.13}$$

The exact scaled normal field is

$$\begin{aligned} &\dot{z} = -\delta S z^3, && \dot{H} = 0, \\ &\dot{A} = A - \tfrac{1}{2}z^2 S + \tfrac{1}{2}\delta z^2\{-\epsilon(1-az) + 2\chi S - SD_{\mathrm{o}}\}, \\ &\dot{\mathsf{B}} = -\mathsf{B} + \tfrac{1}{2}z^2 S + \tfrac{1}{2}\delta z^2\{-\epsilon(1-az) + 2\chi S + SD_{\mathrm{o}}\}. \end{aligned} \tag{3.14}$$

Its positive energy root is smooth through $\delta = 0$. With $C = A + \mathsf{B}$, the normal subsystem there is

$$\dot{C} = D_{\mathrm{o}}, \qquad \dot{D}_{\mathrm{o}} = (1-z^2)C - z^2\chi_0(z),$$

and has rates $\pm\sqrt{1-z^2}$. On $0 \le z \le z_* < 1$, adapted quotient metrics give, for some $\theta > 0$,

$$\mu_2(C_{\tan}) \le \theta, \quad \|B_{\mathrm{cr}}\|, \|D_{\mathrm{cr}}\| \le \theta, \quad a_{\mathrm{n}} \ge \sqrt{1-z_*^2} - \theta, \quad 8\theta < \sqrt{1-z_*^2}. \tag{3.15}$$

The corner itself is regular in

$$s_\epsilon = \sqrt{\epsilon}, \quad \sigma = \sqrt{\delta_1}, \quad \Pi = p_1/\delta_1, \quad \Omega = \frac{1 - v_1 + (s_\epsilon/3)r_1^2}{\delta_1}, \quad r = r_1\sigma. \tag{3.16}$$

On the positive energy root, the exact K_1 field is

$$\begin{aligned} &r_1' = \tfrac{s_\epsilon}{2}\sigma^2\Pi r_1, && \sigma' = -\tfrac{s_\epsilon}{2}\sigma^3\Pi, \quad H' = 0, \\ &\Pi' = \Omega - \tfrac{s_\epsilon}{2}\sigma^2\Pi^2, \\ &\Omega' = (2s_\epsilon + \epsilon r_1^2)\Pi - s_\epsilon q_1 - s_\epsilon\sigma^2\Pi\Omega, \\ &q_1' = \sigma^2\{1 - r_1\sigma^3 a_2 - \tfrac{3}{2}s_\epsilon\Pi q_1\}. \end{aligned} \tag{3.17}$$

Here q_1 is the unique positive root of the regular energy equation; its implicit derivative is $12 s_\epsilon q_1 > 0$. At $\sigma = 0$,

$$q_{10} = \sqrt{\frac{8 + 3 s_\epsilon r_1^2}{6 s_\epsilon}}, \qquad \Pi_0 = \frac{q_{10}}{2 + s_\epsilon r_1^2}, \qquad \Omega_0 = 0,$$

and the normal rates are $\pm\lambda_1$, where

$$\lambda_1 = \sqrt{s_\epsilon (2 + s_\epsilon r_1^2)} \geq \lambda_* := \sqrt{2}\, \epsilon_-^{1/4}.$$

On the fixed cylinder $0 \leq r_1 \leq R$, $0 \leq \sigma \leq \sigma_0$, the collars can therefore be chosen so that the order-j quotient gaps obey

$$\gamma_j \geq \lambda_* - (2j + 2)\theta > 0, \qquad 0 \leq j \leq 3. \tag{3.18}$$

The exact transition to (3.13) is

$$\begin{aligned} z &= (1 + s_\epsilon r_1^2)^{-1}, \qquad & \chi &= z^2 \epsilon^{3/2} r_1^3 q_1, \\ A + \mathsf{B} &= \epsilon r_1 \Pi - \chi, \qquad & A - \mathsf{B} &= \epsilon z r_1^2 \Omega. \end{aligned} \tag{3.19}$$

The invariant identity $(r_1 \sigma)' = 0$ and $\Pi_0 > 0$ give a directed base crossing from the K_2 overlap to the outer overlap. The relative overflowing invariant-manifold theorem applied on doubled collars, followed by the common-cone gluing allowed by (3.15)–(3.18), constructs the codimension-one graph through the block with the mixed jets stated below.

Proposition 3.2 (Central–outer matching)**.** *Under Hypothesis 1.1, after one final choice of the positive annulus, a fixed patch of* (3.8) *has a unique backward saturation through the outer tube, resolved K_1 corner, and K_2 overlap. Its trace on a fixed central section is a cooriented C^3 graph*

$$\mathcal{W}^{\text{match}}_{\text{out},\mu}. \tag{3.20}$$

If G_0 is the frozen singular comparison graph and G_μ the resolved trace, then

$$\|G_\mu - G_0\|_{C^2} \leq Cr, \qquad \sup_{i+j \leq 3,\ j \leq 2} \left\| D_Z^i D_\mu^j G_\mu \right\| < \infty. \tag{3.21}$$

Under $(\mathrm{H}_{\text{Jost}})$*, the endpoint exchange pairing satisfies*

$$\chi_{\text{ex},\mu} = 144\sqrt{3} + O(r) \geq 72\sqrt{3}. \tag{3.22}$$

This is distinct from the source matching determinant. If $y_\mu(\phi, t) = \Phi_\mu^t(S_\mu(\phi))$ and $\overline{g}_{\text{c},\mu}$ defines the matched target on $h_\text{c} = 0$, choose $N_\text{c} \in T(h_\text{c} = 0)$ with $Dg_\text{a}(N_\text{c}) = 1$. Its projected row is nonzero on N_c after the final annulus choice. Rescale $\overline{g}_{\text{c},\mu}$ so that this value is one, and let $\widetilde{\overline{g}}_{\text{c},\mu}$ be any flowbox extension. Its normalized intrinsic section row is

$$\overline{L}_{\text{c},\mu} = d\widetilde{\overline{g}}_{\text{c},\mu} - \frac{d\widetilde{\overline{g}}_{\text{c},\mu}(F_\mu)}{dh_\text{c}(F_\mu)}\, dh_\text{c}, \qquad \overline{L}_{\text{c},\mu}(N_\text{c}) = 1. \tag{3.23}$$

On the section this row is independent of the chosen extension, annihilates F_μ, and satisfies $\overline{L}_{c,\mu} = Dg_a + O_{C^1}(r)$. Put

$$\mathfrak{M}_\mu(\phi, t) = \begin{pmatrix} \widetilde{\overline{g}}_{c,\mu}(y_\mu(\phi, t)) \\ h_c(y_\mu(\phi, t)) \end{pmatrix}. \tag{3.24}$$

At a zero, exact row reduction gives

$$\det D_{(\phi,t)}\mathfrak{M}_\mu = s_\mu \chi_\mu, \qquad s_\mu = dh_c(F_\mu), \quad \chi_\mu = \overline{L}_{c,\mu} D_\phi y_\mu. \tag{3.25}$$

The continuations of (1.9) satisfy $|s_\mu| \geq s_* > 0$ and $|\chi_\mu - \chi_0| < |\chi_0|/2$. Thus the phase/time operator is uniformly invertible and selects a unique C^2 source phase whose orbit crosses the original finite gate and stays on (3.20) to the algebraic end. Every truncated matching action satisfies exact moving-cut covariance.

Proof. The scaled outer graph extends to the comparison face $r = 0$. On a resolved K_1 block, uniform cone and bunching estimates give an overflowing center graph with fixed-extension uniqueness. Exact chart transitions transport it to both neighboring charts, and uniqueness makes the graphs agree on the overlaps. The final central trace is therefore an actual saturation of the same future-staying hypersurface, not a second asymptotic sheet. The resolved field and chart embeddings are C^2 in μ on the closed comparison family. The order-j quotient gaps through $j = 3$ give $G \in C^0_\mu C^3_Z \cap C^1_\mu C^2_Z \cap C^2_\mu C^1_Z$, precisely the mixed-order-three bound in (3.21); the graph-transform also gives its $O(r)$ statement without a boundary trace loss.

The endpoint-normalized adjoint row is transported through this same atlas. At $r = 0$, $(\mathrm{H}_{\mathrm{Jost}})$ gives (1.10); continuity gives (3.22), so the directed endpoint conormal does not collapse. Separately, the normalized central target row converges to Dg_a, and the source flight converges in C^1. Therefore (1.9) gives the two strict factors in (3.25). The parameter implicit-function theorem selects the source phase and flight time. Strict central first-hit margins keep the selected orbit in the same gate component; invariance and local maximality then keep it on the future sheet.

Finally, all chart primitives are pullbacks of the single physical form λ_δ. For a first-hit map $P : C_0 \to C_1$ with flight time T_P, the ambient identity is

$$P^*\lambda_\delta - \lambda_\delta = d\mathcal{A}_P + T_P\, dH_{\mathrm{ph}}. \tag{3.26}$$

On zero energy the final term vanishes on tangent vectors. Splitting the same finite orbit integral proves exact composition and moving-cut covariance. □

3.3. **Reference-tail finite parts.** Subtracting finitely many formal asymptotic terms would make the normalization depend on the truncation order. We instead compare every algebraic tail with one complete reference orbit selected by the same vector field. Contraction in the coordinate below makes the difference of both length and action densities integrable.

Fix a small outer cut $z = z_*$ on the matched sheet and use $Q = z^{-2}$. On zero energy, write the stable graph label as b and let $b_{\mathrm{ref}}(Q; \mu)$ be the reference orbit selected by $b(z_*^{-2}) = 0$. Abbreviate, at $z = Q^{-1/2}$,

$$\Gamma = \Gamma_\mu(z, 0, b), \quad \mathrm{Chi} = \mathrm{Chi}(z, 0, b, \Gamma; \mu), \quad \Pi = \delta\,\mathrm{Chi} + \Gamma + b, \quad W = \Gamma - b.$$

Then $dQ/d\tau = 2\Pi$ and the exact scalar reduced equation is

$$\frac{db}{dQ} = G(Q, b; \mu) = \frac{1}{2\Pi}\left[-b + \frac{z^2}{2}\{-\delta^2\epsilon(1 - az) + 2\delta\,\mathrm{Chi}\,\Pi + \Pi + \Pi W\}\right]. \tag{3.27}$$

In particular,

$$\partial_b G(\infty, 0; \mu) = -\frac{1}{2\delta q_*(\epsilon)} < 0. \tag{3.28}$$

Hence any actual tail and the reference tail at the same Q differ exponentially:

$$\left| D_{b_0}^i D_\mu^j \{b(Q; b_0, \mu) - b_{\mathrm{ref}}(Q; \mu)\}\right| \le Ce^{-\eta(Q - Q_*)}, \qquad i + j \le 3,\ j \le 2. \tag{3.29}$$

The physical length and action densities in this coordinate are exactly

$$\begin{aligned} \mathcal{T}(Q, b; \mu) &= \frac{\delta}{2\Pi(Q, b; \mu)} Q^{-1/2}, \\ \mathcal{A}(Q, b; \mu) &= -\frac{\mathrm{Chi}(Q, b; \mu)^2}{2\Pi(Q, b; \mu)} Q^{3/2} + \frac{\epsilon\Pi(Q, b; \mu)}{2} Q^{-1/2}, \end{aligned} \tag{3.30}$$

where Π and Chi are the positive outer momentum and energy root.

Proposition 3.3 (Algebraic finite parts)**.** *Under Hypothesis 1.1, the complete reference integrals*

$$C_{\mathsf{x},\mu}(Q) = \int_{Q_*}^{Q} \mathcal{T}(q, b_{\mathrm{ref}}; \mu)\, dq, \qquad C_{\mathsf{A},\mu}(Q) = \int_{Q_*}^{Q} \mathcal{A}(q, b_{\mathrm{ref}}; \mu)\, dq \tag{3.31}$$

have the leading behavior (1.14)*. The differences between an actual tail and the reference tail are integrable after every mixed derivative of total order at most two, and in fact after one spare state derivative. Consequently* (1.13) *exists, composes with the whole matched arrival patch, and satisfies exact cut, admissible coordinate, reference, and exact-gauge covariance.*

Proof. The negative rate (3.28) gives a contraction on an exponentially weighted half-line. Differentiating its fixed-point equation three times gives (3.29). Since Π is uniformly positive, the mean-value formula applied to (3.30) gives integrable bounds of the form

$$\begin{aligned} |D^{\le 3}(\mathcal{T} - \mathcal{T}_{\mathrm{ref}})| &\le CQ^{-1/2}e^{-\eta(Q-Q_*)}, \\ |D^{\le 3}(\mathcal{A} - \mathcal{A}_{\mathrm{ref}})| &\le C(1 + Q)^{3/2}e^{-\eta(Q-Q_*)}. \end{aligned} \tag{3.32}$$

Dominated convergence proves the finite parts and their derivatives. On the reference orbit, $\Pi = \delta q_* + O(Q^{-1/2})$ and $\mathrm{Chi} = q_* + O(Q^{-1/2})$; inserting these in (3.30) and integrating proves (1.14). Because (3.31) uses the complete field-dependent tail, all lower powers and logarithms are subtracted automatically; no finite Taylor polynomial is being renamed a finite part.

For every finite Q, ordinary spatial-length and line-integral additivity holds before subtraction. The same single reference term appears on both sides, so the equality survives as $Q \to \infty$. Event-free section slides and exact changes $\lambda_\delta \mapsto \lambda_\delta + d\psi$ add endpoint coboundaries. Exponential flatness makes admissible coordinate and reference changes converge to finite endpoint corrections. This proves all stated covariances. □

Both noncompact outcomes now carry mixed-C^2 terminal potentials with exact cut covariance. These are the final inputs needed to turn the compact first-hit partition into a physical return–exit relation.

4. Exhaustive first events and stationary spatial patterns

4.1. **Completion of the physical first-event relation.** The compact central theorem classifies first hits only up to two finite gates. Propositions 2.2 and 3.2 identify the future of each gate. Because both replacements take place inside protected flow boxes with strict ordering margins, they preserve the existing first-event partition rather than introducing new competing hit times.

Concretely, the transverse algebraic carrier is subdivided by the internal label selecting the matched sheet of Proposition 3.2; the carrier keeps its original hit time. Inside the protected pole outcome, an event-free slide replaces the old gate by the actual $x = 10$ carrier in (2.7), which is partitioned into the pole aperture and its complement. The supplement gives the stable proof-dossier location of the full face and competing-time inventory. In the labels below, out and rbox denote the two named finite lateral boundaries.

Proposition 4.1 (Compact-family first events)**.** *Assume Hypothesis 1.1 and the constructions of Propositions 2.1, 2.2, 3.2, and 3.3. Fix $\varkappa$ satisfying (1.17). There are a finite marked cover $V_i \Subset U_i$ of (1.11), an intrinsic residence threshold T_*, an integer N, and local signed source collars $\Sigma_{N,i}$ with the following properties.*

In each marking, $\Sigma_{N,i}$ is the disjoint union

$$\begin{aligned}\Sigma_{N,i} = \bigsqcup_{\substack{\sigma,\sigma'\in\{+,-\}\\ n\geq N}} \mathcal{V}_{\sigma,n}^{\sigma'} \sqcup \bigsqcup_{\substack{\sigma\in\{+,-\}\\ n\geq N}} \mathcal{I}_{\sigma,n} \\ \sqcup \bigsqcup_{\substack{\mathrm{t}\in\{\mathrm{a,p,out,rbox}\}\\ \sigma\in\{+,-\},\ n\geq N,\ \ell}} E_{\mathrm{t},\sigma,n,\ell},\end{aligned} \tag{4.1}$$

where every union ranges only over nonempty connected components. Every point has one local winding, current sign, component label, and first event; every return has one next sign. The stable cut $\mathcal{I}_{\sigma,n}$ is an exit face and is not counted as either adjacent return. For each current sign, both target-sign returns are nonempty for every $n \geq N$. The selected anchored pole and algebraic components, with their fixed source-sign and component labels, are nonempty for every $n \geq N$.

The displayed identity is a stratified disjoint union of connected relative interiors, not a union of open sets of one dimension. In particular, an algebraic terminal stratum is the trace of a cooriented hypersurface, whereas the pole aperture has a top-dimensional relative interior. Their common boundaries occur as separately named lower-dimensional strata.

For $a_* = (\sigma, n, \sigma')$, the completed return map has a cross form

$$x' = f_{a_*}(x, y'), \qquad y = g_{a_*}(x, y') \tag{4.2}$$

on fixed rectangles. Its contraction is uniformly less than one, and for each sign pair there are limiting cross forms such that

$$\|f_{a_*} - f_{\mu,\sigma,\sigma',\infty}\|_{C^2_{z,\mu}} + \|\widetilde{g}_{a_*} - \widetilde{g}_{\mu,\sigma,\sigma',\infty}\|_{C^2_{z,\mu}} \leq C(1+n)^3 e^{-\varkappa n}, \tag{4.3}$$

where

$$\widetilde{g}_{a_*} = g_{a_*}/\varepsilon_{\mu,a_*}, \qquad \varepsilon_{\mu,a_*} = \nu_{N,i}^{-1} \exp\left(-\frac{2\pi\alpha_\mu}{\beta_\mu} n\right). \tag{4.4}$$

On overlaps, the chartwise partitions have a finite common refinement and describe the same physical first-event relation on Σ_{T_*}; their raw winding labels differ at most by bounded integer recoding.

Proof. The transported pole carrier and the pre-existing algebraic carrier are compact transverse physical first hits with strict speeds, earlier-event exclusion, flow-domain buffers, and anchor-to-boundary margins. The algebraic internal label is instead a cooriented subdivision of its carrier: Proposition 3.2 gives its nonzero pulled-back conormal, and it is assigned no new hit time. The pole aperture is a clean product-face subdivision inside the protected $x = 10$ carrier. Thus the old finite list of competing hit times is unchanged. Add the new conormals to the old clean event table and take one finite minimum of all normalized nonzero ranks and strict gaps. Compact first-hit stability preserves the incidence complex and physical order; near a genuine tie the nonzero tie conormal and the fixed priority replace any false time-gap assertion. The old finite block was already exhaustive, and the two operations only subdivide its algebraic and pole outcomes. Hence there is no residual component.

We now verify directly that the compact family meets the input–output contract in $(\mathrm{H}_{\mathrm{ret}})$. On each U_i, Proposition 2.1 provides the exact transverse-action passage and the mixed two-jet expansions (2.5). Its continued homoclinic transversality, together with the frozen matching clause recorded in $(\mathrm{H}_{\mathrm{ret}})$, provides the fixed limiting matching rectangle, inverse, and range margins. Propositions 2.2 and 3.3 provide terminal potentials with one spare state derivative, while Proposition 3.2 supplies the required transverse terminal conormal.

Let $\mathfrak{h}_{\mu,\sigma}$ denote the finite family of defining rows for all physical event faces, internal labels, and pairwise time differences. It satisfies

$$\sup_{i+j\leq 3,\ j\leq 2} \left\| D_Z^i D_\mu^j \mathfrak{h}_{\mu,\sigma} \right\| < \infty. \tag{4.5}$$

For the old rows this follows from finite-flow state-C^3, parameter-C^2 regularity; the algebraic label is supplied by Proposition 3.2, and the pole rows are product faces in the pole chart. On overlaps the rows arise from the same physical flow, and the section changes have uniformly bounded mixed jets. The compact finite cover therefore supplies common constants. Terminal potentials are not event rows; their separate spare-derivative bounds were established in Propositions 2.2 and 3.3. Together with the strict margins in the first paragraph, these are precisely the inputs of $(\mathrm{H}_{\mathrm{ret}})$, now checked on a compact positive-parameter family rather than merely at its singular reference member.

Apply the compact-family conclusion of $(\mathrm{H}_{\mathrm{ret}})$ on each U_i and refine the cover so that there is a common rate

$$\varkappa < \varkappa_i^+ < \frac{2\pi \inf_{\mu\in U_i} \alpha_\mu}{\sup_{\mu\in U_i} \beta_\mu}. \tag{4.6}$$

This order of infimum and supremum is what supplies a uniform passage contraction on the cover member. Inverting the phase law in (2.5) gives

$$|\nu_{\mu,\sigma,n}| = O\left(\exp\left\{-\frac{2\pi\alpha_\mu}{\beta_\mu} n\right\}\right).$$

Each of two parameter differentiations contributes at most a polynomial factor in n; composing with the event rows and completed cross coordinates uses the spare state derivative and yields the uniform bound $C(1+n)^3 e^{-\varkappa n}$. The winding selector and the clean first-event isotopy therefore give (4.1)–(4.3). Because the pole and algebraic anchors lie in fixed component interiors, their selected components persist for every sufficiently large winding.

Finally, residence time and winding differ by a uniformly bounded amount after multiplication by $2\pi/\beta_\mu$. Choose the final N so that every displayed branch lies in the intrinsic collar Σ_{T_*}. On marking overlaps, exact section changes preserve the physical event and change the angular lift by a bounded integer. Equality of the physical first-hit maps on overlaps, followed by a finite common refinement, gives the marked-atlas descent. Thus the physical relation is intrinsic even though there need be neither a global exact chart nor a global integer alphabet. $\square$

4.2. **Length, action, and exact composition.** For a return branch, write

$$\iota_{\mu,a_*}(x,y') = (x, g_{a_*}(x,y'))$$

for the source inclusion. The physical clock factor in (2.4) and the saddle passage give

$$L_{a_*} = \frac{2\pi r}{\epsilon^{1/4}\beta_\mu} n + \widehat{L}_{a_*}, \qquad \mathcal{A}_{a_*} = \int_{a_*} \lambda_\delta, \tag{4.7}$$

and, for each sign pair,

$$\left\|\widehat{L}_{a_*} - \widehat{L}_{\mu,\sigma,\sigma',\infty}\right\|_{C^2_{z,\mu}} + \left\|\mathcal{A}_{a_*} - \mathcal{A}_{\mu,\sigma,\sigma',\infty}\right\|_{C^2_{z,\mu}} \le C(1+n)e^{-\varkappa n}. \tag{4.8}$$

On a terminal branch, composition with the one-spare-derivative pole or algebraic potential gives the rate $C(1+n)^3 e^{-\varkappa n}$ after two derivatives.

For a finite first-hit branch P_γ, the ambient identity (3.26) restricts on zero energy to

$$P_\gamma^*(\lambda_\delta|_{\Sigma_\gamma^+}) - \lambda_\delta|_{\Sigma_\gamma^-} = d\mathcal{A}_\gamma. \tag{4.9}$$

If two branches compose, then

$$\mathcal{A}_{\gamma_2\circ\gamma_1} = \mathcal{A}_{\gamma_1} + \mathcal{A}_{\gamma_2} \circ P_{\gamma_1}. \tag{4.10}$$

At either end, (4.10) is imposed at a finite cutoff; only the terminal segment receives its counterterm; the cutoff limit is taken last. This proves item 3 of Theorem 1.2 and prevents a terminal subtraction from being counted twice.

4.3. **Coding and return to the PDE.**

Completion of the proof of Theorem 1.2. The return rectangles in Proposition 4.1, their full sign adjacency, the completed cross forms, and the strict first-exit partition satisfy $(\mathrm{H}_{\mathrm{ret}})$. The resulting two-sided edge shift on (1.15) is therefore conjugate to the trapped return set. The roof and action potentials have summable variations by (4.3) and (4.8). A contraction on the product of the rectangles for a primitive cyclic word gives one periodic spatial orbit. Summing (4.7) and using the exact cocycle gives (1.18) and the corresponding closed-action formula.

For a finite word $a_0 \cdots a_{m-1}$, use the completed section coordinates in which the selected primary unstable-image trace is $x = 0$ and the local stable trace is $y = 0$. Solve

$$x_{j+1} = f_{a_j}(x_j, y_{j+1}), \qquad y_j = g_{a_j}(x_j, y_{j+1}), \qquad x_0 = 0, \quad y_m = 0. \tag{4.11}$$

The product-box map is a contraction, hence has a unique solution. Internal edges lie in the regular part of the completed sections; the first boundary condition puts the backward orbit on the selected unstable manifold and the last puts the forward orbit on the local stable manifold. The resulting orbit is homoclinic to the homogeneous state and makes the prescribed finite sequence of high-winding visits. Words of arbitrary finite length therefore give multipulse profiles. The usual inverse-limit construction gives every bi-infinite coded orbit; nonperiodic itineraries remain nonperiodic because the first-event itinerary is injective on the trapped set.

All coded orbits remain in the compact saddle block and finite global flight tube, and branch times have a positive lower bound, so no finite spatial-time accumulation occurs. Applying the exact inverse transformation (2.1) gives a complete smooth physical orbit of (1.2). Eliminating p and q yields

$$0 = v - f(u) + du_{\mathsf{xx}}, \qquad 0 = \epsilon(a-u) + v_{\mathsf{xx}}, \tag{4.12}$$

which is the stationary form of (1.1). Finite marked-atlas covariance identifies the physical profile, closed period, and closed action on overlaps, although its raw integer word may be recoded. This proves all remaining items of Theorem 1.2. $\square$

5. Consequences and scope

5.1. **An exact temporal Fourier obstruction.** The existence proof above uses global spatial dynamics and does not identify the resulting profiles with a local classical Turing branch. The following exact calculation separates those two mechanisms at the frozen parameters.

Proposition 5.1 (No classical stationary Turing onset)**.** *Let $d, \epsilon > 0$, let $\alpha = f'(a) = a^2 - 1$, and put $q = k^2$. The homogeneous mode of (1.1) is strictly spectrally stable if and only if $\alpha > 0$; in that case all Fourier modes are strictly stable. A stationary zero eigenvalue at some $k > 0$ exists if and only if*

$$\alpha \leq -2\sqrt{d\epsilon} < 0.$$

Hence (1.1) admits no classical stationary Turing onset from a stable homogeneous equilibrium.

Moreover, throughout the explicit box

$$\frac{1}{25} \leq r \leq \frac{2}{25}, \qquad |a_2| \leq \frac{1}{4}, \qquad \frac{4}{5} \leq \epsilon \leq \frac{6}{5}, \tag{5.1}$$

there is no positive-wavenumber stationary zero. When $2r^2\sqrt{\epsilon} < 1$, the two stationary-neutral branches are

$$a_{2,\mathrm{T}}^{\pm}(r,\epsilon) = \frac{\pm\sqrt{1 - 2r^2\sqrt{\epsilon}} - 1}{\sqrt{\epsilon}\, r^3}, \qquad k_c = \frac{\epsilon^{1/4}}{r}. \tag{5.2}$$

The positive-fold branch is $a_{2,\mathrm{T}}^{+}$; throughout (5.1) it satisfies $a_{2,\mathrm{T}}^{+} < -1/r \leq -25/2$, while the minus branch is still more negative. Thus neither branch meets the frozen a_2 interval, and both lie on a homogeneously unstable background.

Proof. The Fourier symbol at $(a, f(a))$ is

$$L(k) = \begin{pmatrix} -\alpha - dq & 1 \\ -\epsilon & -q \end{pmatrix}, \qquad \operatorname{tr} L = -\alpha - (1+d)q, \qquad \det L = dq^2 + \alpha q + \epsilon.$$

At $q = 0$, the Routh–Hurwitz criterion gives strict stability precisely for $\alpha > 0$, which also makes the trace negative and determinant positive for every $q \geq 0$. A stationary zero at $q > 0$ is a zero of the convex quadratic $dq^2 + \alpha q + \epsilon$. Its minimizer is positive and its minimum is nonpositive precisely when $\alpha \leq -2\sqrt{d\epsilon}$. If $\alpha = 0$, the zero mode is $\lambda = \pm i\sqrt{\epsilon}$, while every $q > 0$ is strictly stable. If $\alpha < 0$, the zero mode is already unstable, so any finite-q stationary zero occurs on an unstable homogeneous background rather than at a classical onset.

In the cusp coordinates (1.7),

$$\alpha + 2r^2\sqrt{\epsilon} = \sqrt{\epsilon}\, r^2 \left(2 + 2ra_2 + \sqrt{\epsilon}\, r^4 a_2^2\right),$$

and the bracket in (5.1) is at least $2 - 1/25 = 49/25$. This proves the strict box separation. Equality in the quadratic criterion has $q_c = \sqrt{\epsilon}/r^2$; solving it for

a_2 gives (5.2). The branch estimate follows from $\sqrt{1-y} < 1-y/2$ for $0 < y < 1$, which applies throughout the box because $2r^2\sqrt{\epsilon} \le (8/625)\sqrt{6/5} < 1$. □

Remark 5.2 (Terminology and claim boundary). Vo, Doelman, and Kaper call the corresponding spatial reversible 1:1 curve a singular Turing bifurcation [8]. The proposition uses the narrower classical temporal definition in which diffusion destabilizes a homogeneous state with stable reaction kinetics. The terms are compatible: on the spatial 1:1 curve here, the temporal homogeneous mode is already unstable. The proposition does not exclude a branch born outside (5.1) and globally continued into it, finite-wavenumber growth on an already unstable background, an oscillatory or nonlinear mechanism, wavelength selection, or temporal selection of a nonconstant profile.

The model-specific contribution is to continue two finite singular gates in the full positive-diffusion van der Pol system. The invariant pole cone and regular-singular analysis construct the finite-distance end; the outer staying graph and resolved bridge construct the infinite-distance end. Their finite parts supply the terminal potentials needed to turn the abstract first-hit block into an exhaustive physical return–exit relation.

The pole and algebraic ends are two distinguished exit components; the stable cut and named lateral exits are the others. No exit branch gives a bounded profile on the whole spatial line. The bi-infinite trapped relation yields periodic and bounded aperiodic profiles, while its boundary-completed finite words yield multipulses. Thus first-event exhaustiveness is not confused with the mechanism producing stationary patterns.

The stationary existence result remains conditional on Hypothesis 1.1, local to a sufficiently high-winding source collar, and uniform only on the existential annulus (1.11). It gives neither an explicit validated parameter box nor uniformity at $r = 0$. The independent variable here is the spatial coordinate x: spatial entropy or an aperiodic stationary profile is not temporal chaos for the parabolic PDE. Temporal spectral or nonlinear stability, nonlinear temporal selection, canard tracking, and experimental observability remain separate questions.

Data and supplementary material. The fixed-system theorem and the application evidence release are public immutable records. The supplement maps each imported or application claim to those records, the versioned proof dossiers, certificates, hashes, environment, and replay commands; it also retains the V1–V7 crosswalk outside the mathematical narrative.

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